literature review

Create a Literature Review Matrix

No.	Reference	Problem Addressed	Dataset	Methodology	Results	Limitations	Key Contribution
1	Author et al., 2025	Disease classification	PTB-XL	CNN	95% Accuracy	Small dataset	Automated feature extraction
2	Author et al., 2024	Risk prediction	MIT-BIH	CNN-LSTM	96% F1-score	Binary classification only	Temporal modeling

Identify Common Themes

After filling the table, look for patterns. For example,

Group A: CNN-Based Methods

- Paper 1
- Paper 3
- Paper 7
- Paper 10

Group B: CNN-LSTM Methods

- Paper 2
- Paper 5
- Paper 8

Group C: Transformer-Based Methods

- Paper 4
- Paper 6
- Paper 9

Group D: Explainable AI Methods

- Paper 11
- Paper 12
- Paper 13

This thematic grouping becomes the structure of your Related Work section.

Compare Instead of Summarizing

Many beginners write:

Smith et al. used CNN and obtained 95% accuracy. Jones et al. used LSTM and obtained 96% accuracy.

This becomes a list of summaries.

Instead write:

Several studies have employed CNN architectures for ECG analysis due to their ability to automatically learn spatial features from raw signals. While CNN-based models achieved promising classification performance, their ability to capture long-term temporal dependencies remains limited. To address this issue, researchers introduced hybrid CNN-LSTM architectures that combine spatial feature extraction with temporal sequence modeling.

Notice that this compares and connects studies.

Find the Research Gap

Now ask the following questions:

Dataset Gap

- Small datasets?
- Single-center datasets?
- Lack of real-world validation?

Methodological Gap

- Only CNN?
- No temporal modeling?
- No multimodal learning?

Explainability Gap

- Black-box predictions?
- No explainable AI?

Application Gap

- Binary classification only?
- No disease progression prediction?
- No risk stratification?

Write every limitation you find.